

Math 521, Section 2
Homework 13
Solutions

Problem 1. Consider the sequence of functions $f_n : [0, 1] \rightarrow \mathbb{R}$ given by

$$f_n(x) = \begin{cases} nx & \text{if } x \in [0, \frac{1}{n}], \\ 2 - nx & \text{if } x \in (\frac{1}{n}, \frac{2}{n}], \\ 0 & \text{if } x \in (\frac{2}{n}, 1]. \end{cases}$$

- (a) Prove that f_n converges pointwise to the function $f : [0, 1] \rightarrow \mathbb{R}$, $f(x) = 0$.
- (b) Prove that f_n does not converge uniformly to any function $g : [0, 1] \rightarrow \mathbb{R}$.

Solution: (a) Fix $x \in [0, 1]$. We want to show that the sequence of numbers $f_n(x)$ converges to $f(x) = 0$.

- Case: $x = 0$. Then $f_n(0) = 0$ for all n , and so $f_n(0) \rightarrow 0$.
- Case: $x \in (0, 1]$. Then there exists an $N \in \mathbb{N}$ so that $\frac{2}{N} < x$. For $n \geq N$ we have $\frac{2}{n} \leq \frac{2}{N} < x$, and so $f_n(x) = 0$.

(b) Suppose towards a contradiction that f_n converges uniformly to some function $g : [0, 1] \rightarrow \mathbb{R}$. Then f_n also converges pointwise to g , and so by part (a) we have $g(x) = 0$ for all $x \in [0, 1]$.

Consider $\epsilon = 1$. As f_n converges uniformly to g , there must be some N so that

$$|f_n(x) - g(x)| < 1 \quad \text{for all } n \geq N \text{ and } x \in [0, 1].$$

However, for $x = \frac{1}{n}$ we have

$$|f_n(\frac{1}{n}) - g(\frac{1}{n})| = |1 - 0| = 1$$

for any $n \geq N$, which is a contradiction. □

Problem 2. Consider the series of functions $\sum_{n=0}^{\infty} \frac{x^n}{1+x^n}$.

- (a) Prove that for any $r \in (0, 1)$, the series converges uniformly on $[0, r]$.
- (b) Prove that the series does not converge uniformly on $[0, 1)$.

Solution: (a) Fix $r \in (0, 1)$. Then for $x \in [0, r]$ we have

$$\left| \frac{x^n}{1+x^n} \right| \leq \frac{r^n}{1+0} = r^n.$$

Note that the geometric series $\sum r^n$ converges since $r < 1$. Therefore, $\sum \frac{x^n}{1+x^n}$ converges uniformly on $[0, r]$ by the Weierstrass M -test.

(b) Let $s_n(x) = \sum_{k=0}^n \frac{x^k}{1+x^k}$ denote the n th partial sum. Note that $s_n \in C([0, 1))$, since each function $\frac{x^k}{1+x^k}$ is continuous on the larger interval $[0, 1]$, and thus is continuous and bounded on $[0, 1)$ by the extreme value theorem. On the other hand,

$$d_{\text{sup}}(s_n, s_{n-1}) = \sup_{x \in [0, 1)} \left| \frac{x^n}{1+x^n} \right| = \frac{1}{2}$$

for any $n \geq 1$. (In the last equality, we noted that the function $f(x) = \frac{x^n}{1+x^n}$ is increasing on $[0, 1)$ since $f'(x) = nx^{n-1}(1+x^n)^{-2} \geq 0$ for $x \geq 0$, and so the supremum of f must occur at $x = 1$.) Therefore, the sequence $\{s_n\}_{n \geq 0}$ is not Cauchy in $C([0, 1))$, and so cannot be converging uniformly. $\square$

Problem 3. Consider the power series $\sum_{n=1}^{\infty} \frac{x^n}{\sqrt{n}}$.

- (a) Find its radius of convergence R .
- (b) Determine if the series converges at each of the endpoints $x = R$ and $x = -R$.

Solution: (a) Let $c_n = \frac{1}{\sqrt{n}}$. Then

$$|c_n|^{1/n} = \frac{1}{\sqrt[n]{n}} \longrightarrow 1$$

as $n \rightarrow \infty$ (by the limit law for division, HW4#5, and Lecture 15). Therefore $\limsup |c_n|^{1/n} = 1$, and so $R = 1$.

(b) At $x = 1$, we have the series $\sum \frac{1}{\sqrt{n}}$. This diverges by Lecture 16.

At $x = -1$, we have the series $\sum \frac{(-1)^n}{\sqrt{n}}$. This converges by the alternating series test, since $\frac{1}{\sqrt{n}} \rightarrow 0$ as $n \rightarrow \infty$.

Altogether, we conclude that the power series converges if and only if $x \in [-1, 1)$. $\square$

Problem 4. For each of the following power series, determine the radius of convergence.

$$(a) \sum_{n=0}^{\infty} n^2 x^n \qquad (b) \sum_{n=1}^{\infty} \left(\frac{x}{n}\right)^n \qquad (c) \sum_{n=0}^{\infty} n! x^n$$

Solution: (a) Let $c_n = n^2$. Then

$$|c_n|^{1/n} = (n^{1/n})^2 \longrightarrow 1$$

as $n \rightarrow \infty$ (by the limit law for multiplication and Lecture 15). Therefore $\limsup |c_n|^{1/n} = 1$, and so $R = 1$.

(b) Let $c_n = n^{-n}$. Then

$$|c_n|^{1/n} = \frac{1}{n} \longrightarrow 0$$

as $n \rightarrow \infty$. Therefore $\limsup |c_n|^{1/n} = 0$, and so $R = \infty$.

(c) Let $c_n = n!$. Then

$$|c_n|^{1/n} = (n!)^{1/n} \longrightarrow \infty$$

as $n \rightarrow \infty$ (by Lecture 15). Therefore $\limsup |c_n|^{1/n} = \infty$, and so $R = 0$. □

Problem 5. The *hyperbolic sine* function is defined by $\sinh x = \frac{1}{2}(e^x - e^{-x})$.

- (a) Find its Taylor series centered at $x = 0$.
- (b) Prove that this series converges to $\sinh x$ for any $x \in \mathbb{R}$.

Solution: (a) Set $f(x) = \sinh x$. As $\frac{d}{dx}e^x = e^x$, we have $\frac{d}{dx}e^{-x} = -e^{-x}$. Therefore

$$f^{(n)}(x) = \begin{cases} \frac{1}{2}(e^x - e^{-x}) & \text{if } n \text{ is even,} \\ \frac{1}{2}(e^x + e^{-x}) & \text{if } n \text{ is odd,} \end{cases}$$

and so

$$f^{(n)}(0) = \begin{cases} 0 & \text{if } n \text{ is even,} \\ 1 & \text{if } n \text{ is odd.} \end{cases}$$

So the Taylor series of $\sinh x$ centered at $x = 0$ is

$$\sum_{k=0}^{\infty} \frac{x^{2k+1}}{(2k+1)!}.$$

- (b) Fix $x_0 \in \mathbb{R}$. Set $r = |x_0| + 1$, so that $x_0 \in (-r, r)$. Then

$$|f^{(n)}(x)| = \left| \frac{1}{2}(e^x - (-1)^n e^{-x}) \right| \leq \frac{1}{2}(e^x + e^{-x}) \leq \frac{1}{2}(e^r + e^r) = e^r \quad \text{for all } x \in (-r, r),$$

for any $n \geq 0$. Therefore, by Lecture 36, we have

$$\sinh x = \sum_{k=0}^{\infty} \frac{x^{2k+1}}{(2k+1)!}$$

for any $x \in (-r, r)$. In particular, equality holds for the point $x = x_0$ we started with. As $x_0 \in \mathbb{R}$ was arbitrary, we conclude that equality is true for any $x_0 \in \mathbb{R}$. $\square$

Problem 6. Consider the function $f : (-1, \infty) \rightarrow \mathbb{R}$, $f(x) = \log(1 + x)$.

- (a) Using induction, prove that $f^{(n)}(x) = (-1)^{n+1}(n-1)!(1+x)^{-n}$ for all $n \geq 1$.
- (b) Using part (a), find the Taylor series of f centered at $x = 0$.
- (c) Suppose we want to approximate the number $f(1) = \log 2$. Prove that for any $n \geq 2$,

$$\left| f(1) - \sum_{k=1}^{n-1} \frac{(-1)^{k+1}}{k} \right| \leq \frac{1}{n}.$$

This gives us a way to approximate $\log 2$ and to guarantee that the approximation is as accurate as we want, without having to know the value of $\log 2$ beforehand.

Solution: (a) Base case: For $n = 1$, we have $f'(x) = (1+x)^{-1}$ by the chain rule.
Inductive hypothesis: Assume that

$$f^{(n)}(x) = \frac{(-1)^{n+1}(n-1)!}{(1+x)^n}$$

for some $n \geq 1$. Then, by the chain rule we have

$$f^{(n+1)}(x) = (-n) \frac{(-1)^{n+1}(n-1)!}{(1+x)^{n+1}} = \frac{(-1)^{n+2} n!}{(1+x)^{n+1}},$$

and so the statement is true for $n+1$.

Therefore, by induction, we conclude that the formula holds for all $n \in \mathbb{N}$.

(b) By part (a),

$$\frac{f^{(n)}(0)}{n!} = \frac{(-1)^{n+1}(n-1)!}{n!} = \frac{(-1)^{n+1}}{n}$$

for any $n \geq 1$. When $n = 0$, we have $f(0) = 0$. Therefore the Taylor series of f centered at $x = 0$ is

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n.$$

(c) Fix $n \geq 2$. By Taylor's theorem with remainder, there exists a point $y \in (0, 1)$ so that

$$f(1) = \sum_{k=1}^{n-1} \frac{(-1)^{k+1}}{k} (1-0)^k + \frac{f^{(n)}(y)}{n!} (1-0)^n.$$

Rearranging this expression and using part (a), we find

$$\left| f(1) - \sum_{k=1}^{n-1} \frac{(-1)^{k+1}}{k} \right| = \frac{|f^{(n)}(y)|}{n!} = \frac{1}{n|1+y|^n} \leq \frac{1}{n}.$$

In the last step, we noted that $y > 0$. □